1. (a) We have

$$2+5+8+\ldots+3n-1=\frac{n(3n+1)}{2}$$

Rewritten using summation gives us

$$\sum_{i=1}^{n} 3i-1=\frac{n(3n+1)}{2}$$

Proof
BASE CASE:

Let $n=1$

$$\sum_{i=1}^{n} 3i-1=\sum_{i=1}^{1} 3(1)-1=3-1$$
$$=\boxed{2}$$

$$\frac{n(3n+1)}{2}=\frac{1\cdot(3(1)+1)}{2}=\frac{3+1}{2}$$
$$=\frac{4}{2}$$
$$=\boxed{2}$$

$$2=2$$
$$\therefore n=1 \text{ is true}$$

INDUCTIVE CASE:

Assume that

$$\sum_{i=1}^{n} 3n-1=\frac{n(3n+1)}{2}$$

We must prove

$$\sum_{i=1}^{n+1} 3i-1=\frac{(n+1)(3(n+1)+1)}{2}$$
$$=\frac{(n+1)(3n+4)}{2}$$

$$\begin{aligned}
\sum_{i=1}^{n+1} 3i-1 &= \sum_{i=1}^{n} 3i-1+(3(n+1)-1) \\
&= \frac{n(3n+1)}{2}+3(n+1)-1 \\
&= \frac{n(3n+1)}{2}+3n+2 \\
&= \frac{n(3n+1)}{2}+\frac{6n+4}{2} \\
&= \frac{n(3n+1)+(6n+4)}{2} \\
&= \frac{3n^2+n+6n+4}{2} \\
&= \frac{3n^2+7n+4}{2} \\
&= \frac{(n+\frac{3}{3})(n+\frac{4}{3})}{2} \\
&= \boxed{\frac{(n+1)(3n+4)}{2}}
\end{aligned}$$

$$\frac{(n+1)(3n+4)}{2} = \frac{(n+1)(3n+4)}{2}\checkmark$$

(b) We have

$$\prod_{k=1}^{n}(1+\frac{1}{k}) = n+1$$

Proof
BASE CASE:

Let $n = 1$

$$\begin{aligned}
\prod_{k=1}^{n}(1+\frac{1}{k}) = \prod_{k=1}^{1}(1+\frac{1}{k}) &= 1+\frac{1}{1} \\
&= 1+1 \\
&= \boxed{2}
\end{aligned}$$

$$n+1 = 1+1 = \boxed{2}$$

$$2 = 2$$

$\therefore n = 1$ holds true

INDUCTIVE CASE:

We assume

$$\prod_{k=1}^{n}(1+\frac{1}{k}) = n+1$$

We prove:

$$\prod_{k=1}^{n+1}(1+\frac{1}{k}) = (n+1)+1$$
$$= n+2$$

$$\prod_{k=1}^{n+1}(1+\frac{1}{k}) = \prod_{k=1}^{n}(1+\frac{1}{k})\cdot(1+\frac{1}{n+1})$$
$$= (n+1)\cdot(1+\frac{1}{n+1})$$
$$= (n+1)\cdot\frac{n+1}{n+1}$$
$$= (n+1)+1$$
$$= \boxed{n+2}$$

$$n+2 = n+2\checkmark$$

2. (a) We can split this into

total = 0 numerals + 1 numeral + at least 2 numerals

at least 2 numerals = total - 0 numerals - 1 numeral

Let's start by finding the total. Since there are no restrictions, each position has 36 options

$$total = \underline{36}\cdot\underline{36}\cdot\underline{36}\cdot\underline{36}\cdot\underline{36}\cdot\underline{36}\cdot\underline{36} = 36^7$$

Now for 0 numerals, the restriction is that each character has to be an uppercase letter, so each position has 26 options

$$0 \text{ numerals} = \underline{26}\cdot\underline{26}\cdot\underline{26}\cdot\underline{26}\cdot\underline{26}\cdot\underline{26}\cdot\underline{26} = 26^7$$

Lastly, 1 exact numeral means only one slot has 10 options, but out of 7 slots. So we multiply the count for 1 numeral by the ways we can put that one numeral. The rest of the slots MUST be letters

$$1 \text{ numeral} = \binom{7}{1} \cdot \underline{10} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} \cdot \underline{26} = 7 \cdot 10 \cdot 26^6$$
$$= 70 \cdot 26^6$$

Using our equation from above,

$$\text{at least 2 numerals} = \boxed{6^7 - 26^7 - 70 \cdot 26^6}$$

(b) First, let's find out the number of license plates with MILES being at the start

$$\underbrace{1}_{M} \cdot \underbrace{1}_{I} \cdot \underbrace{1}_{L} \cdot \underbrace{1}_{E} \cdot \underbrace{1}_{S} \cdot 36 \cdot 36 = 36^2$$

Now let's see how many times we can arrange MILES with 7 slots

(a) MILESxx

(b) xMILESx

(c) xxMILES

Since there's three ways to putting MILES into 7 characters, we can just multiple 36^2 by 3 to get

$$\boxed{3 \cdot 36^2}$$

(c) We can think of this as:

(a) choose 3 different characters out of 36

(b) choose 3 different spots of 7 for the first character

(c) choose 3 spots out of 4 for the second character

(d) choose 1 spot out of 1 for the last character

Mathematically, this ends up being

$$\boxed{\binom{36}{3} \cdot \binom{7}{3} \cdot \binom{4}{3} \cdot \binom{1}{1}}$$

(d) Like the last one, we can think of this as

(a) choose 3 different numerals out of 10

(b) choose 1 letter out of 26

(c) choose 3 spots out of 7 to place the numerals

(d) permute the 3 numbers

(e) choose 4 spots of out 4 to place the lettters (basically just put the letters where the numbers aren't)

This ends up being

$$\binom{10}{3} \cdot \binom{26}{1} \cdot \binom{7}{3} \cdot {}_3P_3 \cdot \binom{4}{4}$$
$$= \boxed{\binom{10}{3} \cdot \binom{26}{1} \cdot \binom{7}{3} \cdot 3!}$$

(e) For this one, since alphabetical order means A-Z (1 - 26) and there's 7 letters, the start can only be up to

$$26 - 7 + 1 = 20 \text{ (T)}$$

We can think of the first character essentially controlling the entire string, so in reality all we need to do is choose a valid first letter.

Using the range

$$[1, 20]$$

We have 20 letters to choose from, means our answer is just

$$\boxed{20}$$

3. (a) Since the paintings can all be in the same room, each painting essentially has 12 options to choose from each, so this turns out to be

$$\underbrace{12}_{P1} \cdot \underbrace{12}_{P2} \cdot \ldots \cdot \underbrace{12}_{P9} = \boxed{12^9} \tag{1}$$

(b) In words, the museum curator would essentially be doing:

(a) choose 9 rooms out of 12 for each painting

(b) permute 9 paintings

Mathematically, this is

$$\binom{12}{9} \cdot {}_9P_9 = \boxed{\binom{12}{9} \cdot 9!}$$

(c) We can think of this as the following (exactly 1 painting is in the smallest room)

1 in smallest = 0 in biggest and 1 in smallest + at least 1 in biggest and 1 in smallest

So

at least 1 in biggest and 1 in smallest = 1 in smallest - 0 in biggest and 1 in smallest

For 0 in smallest, each painting just as 11 options (12 - 1 = 11) to exclude the smallest

So we must choose 1 painting out of 9 to place in the smallest, then the rest has 11 choices for rooms

$$1 \text{ in smallest} = \binom{9}{1} \cdot \underbrace{1}_{P1} \cdot \underbrace{11}_{P2} \cdot \underbrace{11}_{P3} \cdot \underbrace{11}_{P4} \cdot \underbrace{11}_{P5} \cdot \underbrace{11}_{P6} \cdot \underbrace{11}_{P7} \cdot \underbrace{11}_{P8} \cdot \underbrace{11}_{P8} = 9 \cdot 11^8$$

For 0 in biggest (with 1 in smallest), one painting must be in the smallest room, and NONE can be in the biggest. This means we must choose one painting out of 9 to be in the smallest, and the rest can be in any of the 10 rooms (can't be in biggest nor smallest, so $12 - 2 = 10$)

$$0 \text{ in biggest and } 1 \text{ in smallest} = \binom{9}{1} \cdot \underbrace{1}_{P1} \cdot \underbrace{10}_{P2} \cdot \underbrace{10}_{P3} \cdot \underbrace{10}_{P4} \cdot \underbrace{10}_{P5} \cdot \underbrace{10}_{P6} \cdot \underbrace{10}_{P7} \cdot \underbrace{10}_{P8} \cdot \underbrace{10}_{P9} = 9 \cdot 10^8$$

So now we can just plug it into our equation

$$\begin{aligned} \text{at least 1 in biggest and 1 in smallest} &= 9 \cdot 11^8 - 9 \cdot 10^8 \\ &= \boxed{9(11^8 - 10^8)} \end{aligned}$$

(d) So our contraint for this one is that A, B, and C must all be in the same room. Because they're all attached to each other, we can just think of all three as one unit, so the total number of paintings we have is essentially 7.

The question then just becomes how many ways can we arrange 7 paintings in 12 rooms?

Just like the other parts, each painting has 12 options for what room to go in

$$\underbrace{12}_{ABC} \cdot \underbrace{12}_{P2} \cdot \underbrace{12}_{P3} \cdot \underbrace{12}_{P4} \cdot \underbrace{12}_{P5} \cdot \underbrace{12}_{P6} \cdot \underbrace{12}_{P7} = \boxed{12^7}$$

(e) Somewhat similar to the last one, we can group the 9 paintings into 3 groups.

Let's define how many ways we can form groups of 3

(a) choose 3 out of 9 to be in one group

(b) choose 3 out of 6 to be in another group

(c) choose 3 out of 3 to be in the last group

So the number arrangements for groups is given by

$$\begin{aligned} \#\text{ of group arrangements} &= \binom{9}{3} \cdot \binom{6}{3} \cdot \binom{3}{3} \\ &= \binom{9}{3} \cdot \binom{6}{3} \end{aligned}$$

so now we have our groups, we can answer the second part of the question

(a) choose 3 out of 12 rooms

Which is just

$$\binom{12}{3}$$

So the total arrangements is just

of ways to arrange 3 even groups from 9 · # of ways to choose 3 rooms out of 12

$$\boxed{\binom{9}{3} \cdot \binom{6}{3} \cdot \binom{12}{3}}$$

4. (a) The number of occurences greater than one for each letter of INCLUSIONEXCLUSION, you get
   - I = 3
   - N = 3
   - C = 2
   - L = 2
   - U = 2
   - S = 2
   - O = 2

   To find the number of arrangements with each duplicate letter not being unique, we can do

$$\#\text{ arrangements without uniqueness} = \frac{\text{total arrangements w/ each letter being unique}}{\#\text{ of permutations of each duplicate letter}}$$

   But another way of thinking about this is just

   (a) choose 3 out of 18 spots to place the I's

   (b) choose 3 out of the 15 spots to place the N's

   (c) choose 2 out of 13 spots to place the C's

   and so on

   which when you simplify this, it's just

$$\frac{18!}{3! \cdot 3! \cdot 2! \cdot 2! \cdot 2! \cdot 2! \cdot 2!} = \boxed{\frac{18!}{(3!)^2 \cdot (2!)^5}}$$

   (b) There are two types of orientation moves:

   (a) roll (3 symmetries)

(b) yaw / pitch (2 symmetries)

This means that there's **6 symmetries** $(3 \cdot 2 = 6)$

So let's just count the number of ways of coloring the prism and ignore the symmetries

There are essentially 9 slots and 9 options. Since they must be different, this is just a **permutation**

$$\begin{aligned} \#\text{ of coloring ways} &= {}_9P_9 \\ &= 9! \end{aligned}$$

We can can just divide the symmetries with the total ways of coloring

$$\boxed{\frac{9!}{6}}$$

(c) With the contraints that order of the games doesn't matter and each player plays only one game, we can just think of this as

(a) choose 6 out of 18 to play one game

(b) choose 6 out of 12 to play another game

(c) choose 6 out of 6 to play the last game

(d) divide by ways of permuting 3 games (since order doesn't matter)

and then for each game, we need to do

(a) choose 3 out of the 6 in the game to be on the first team

(b) choose 3 out of the remaining 3 to be on the other team

(c) divide by ways of permuting 2 teams (since order doesn't matter)

So mathematically this is

$$\begin{aligned} &\frac{\binom{18}{6}\cdot\binom{6}{3}\cdot\binom{3}{3}}{{}_2P_2} \cdot \frac{\binom{12}{6}\cdot\binom{6}{3}\cdot\binom{3}{3}}{{}_2P_2} \cdot \frac{\binom{6}{6}\cdot\binom{6}{3}\cdot\binom{3}{3}}{{}_2P_2} \cdot \frac{1}{{}_3P_3} \\ &= \binom{18}{6}\cdot\binom{12}{6}\cdot\left(\binom{6}{3}\cdot\binom{3}{3}\right)^3 \cdot \binom{6}{6} \cdot \frac{1}{3!\cdot 2!\cdot 2!\cdot 2!} \\ &= \binom{18}{6}\cdot\binom{12}{6}\cdot\binom{6}{3}^3 \cdot \frac{1}{3\cdot 2\cdot 2\cdot 2\cdot 2} \\ &= \boxed{\binom{18}{6}\cdot\binom{12}{6}\cdot\binom{6}{3}^3 \cdot \frac{1}{48}} \end{aligned}$$